

EVERYTHING

Discrete Time Fourier Transform

$$\begin{aligned} X(e^{j\omega}) &= \sum_{n=-\infty}^{\infty} x[n]e^{-j\hat{\omega}n} \\ x[n] &= \frac{1}{2\pi} \int_{\hat{\omega}=2\pi} X(e^{j\hat{\omega}})e^{j\hat{\omega}n}d\hat{\omega} \end{aligned}$$

Discrete Time Fourier Series

$$\begin{aligned} a_k &= \frac{1}{N} \sum_{n=\langle N \rangle} x[n]e^{-jk\hat{\omega}_0n} \\ x[n] &= \sum_{k=\langle N \rangle} a_k e^{jk\hat{\omega}_0n} \end{aligned}$$

Continuous Time Fourier Transform